New York University

TANDON School of Engineering

Department of Mechanical and Aerospace Engineering

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Emergency Drone Rescue System

TarunKumar Palanivelan

Sven Sunny Kottuppallil

Abirami Palaniappan Sirsabesan

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Abstract

The safe landing of drones in emergency situations is a critical challenge. This project proposes a "Emergency Drone Rescue System" that ensures controlled descent and impact mitigation. The system integrates an IMU sensor to detect drone instability, triggering 2 servo motor to deploy a parachute. Additionally, ultrasonic sensors monitor ground distance, activating 4 servo motor that deploys the safety landing gears when the drone is near the ground. A functional drone model incorporating essential flight components has been developed to validate the system.

Introduction

Unmanned Aerial Vehicles (UAVs) have become indispensable in various industries, including surveillance, delivery, agriculture, and disaster management. However, ensuring a safe landing in emergency scenarios remains a significant challenge. Conventional drones lack intelligent mechanisms to prevent damage in cases of sudden failure, making them prone to crashes due to unstable conditions or technical malfunctions. A robust safety system that enables controlled descent and minimizes impact force is crucial for enhancing the reliability of UAVs.

This project introduces a propeller-based drone safety landing system, incorporating an IMU sensor for instability detection, ultrasonic sensors for ground proximity measurement, and a dual-action safety mechanism consisting of a parachute and an safety landing gear. When an imbalance is detected, the parachute deploys automatically, slowing down the descent. Simultaneously, if the drone approaches a critical altitude, then the safety landing gears gets deployed.

The implementation of this system significantly enhances drone safety, particularly in high-risk environments. The project explores the use of lightweight, cost-effective, and efficient electronic components to build a practical and deployable solution. With further refinement, this technology has the potential to revolutionize drone safety, paving the way for more reliable and autonomous UAV operations.

Problem Statement

Drones face critical risks during emergencies such as power failures, strong winds, or component malfunctions. Without an effective landing system, these situations can lead to crashes, resulting in expensive replacements and safety hazards.

Drones are prone to crashes due to motors failures, ESC burnouts and main power supply failure. Although current drones have return to base failsafe, they cannot withstand a Motor or a ESC failure.

Traditional safety mechanisms like parachutes alone may not provide complete protection.

Fig 1: ESC Failure

Fig 2: Motor Burn

Thus, an integrated system incorporating multiple sensors and actuators is needed to ensure a safer landing.

Proposed Solution

The proposed system consists of an IMU sensor, ultrasonic sensors, parachute deployment via a servo motor, and a safety landing gear. The IMU sensor detects imbalance and activates the parachute. If the drone is within 200 cm of the ground, the ultrasonic sensors trigger 4 servo motor placed on the bottom of the drone which deploys the safety landing gears. This multi-layered approach enhances landing safety.

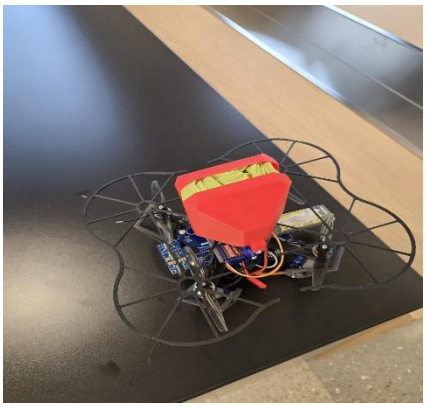


Fig 3: Prototype- Top View

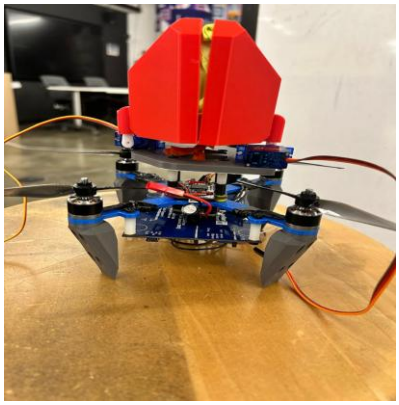


Fig 4: Prototype- Side View

Components Selection

- **IMU Sensor**: Detects instability.
- **Ultrasonic Sensors**: Measures ground and surrounding distances.
- **Parallax Propeller Board**: Microcontroller for processing data.
- **Servo Motors (x6)**: Controls parachute and CO2 canister release.
- **Parachute:** Placed on top of the drone
- **Safety Landing Gears (3D):** Spring suspended design to withstand the exerted force
- **Drone Components**: Frame, flight controller, ESC, battery, motors, propellers, transmitter-receiver.

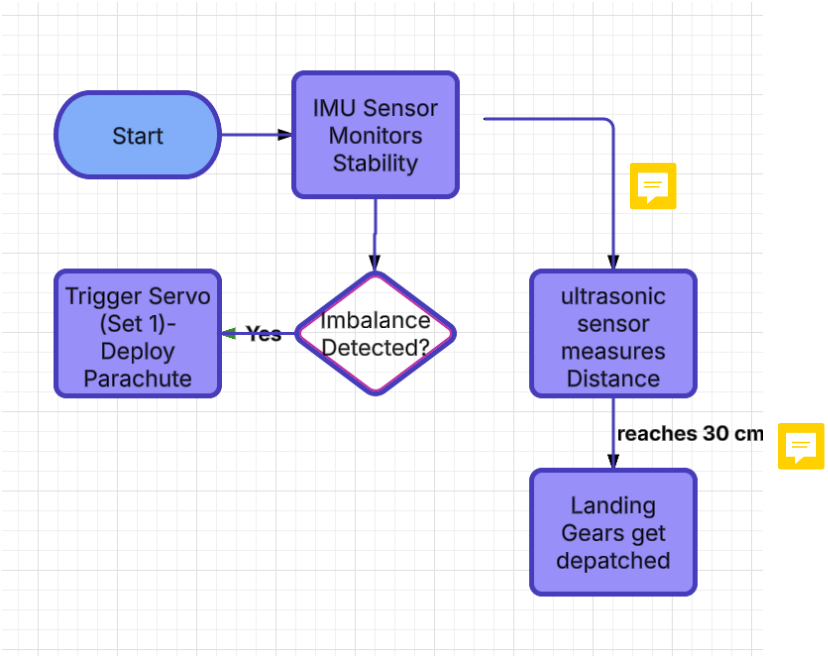
Working Methodology

Based on a multi-layered approach that ensures a controlled and safe landing in case of an emergency. The system primarily relies on an **IMU sensor (MPU6050)** to detect sudden instability or unusual tilting of the drone. When an imbalance is detected, the propeller triggers 2 **servo motor (MG995)** that releases a **parachute**, slowing down the descent. Meanwhile, **ultrasonic sensors (HC-SR04)** continuously measure the distance between the drone and the ground. The sensors are strategically placed downward to ensure all possible landing scenarios are covered. If the drone reaches a critical **altitude of 200 cm**, the 4 **servo motor** is activated, triggering the **Safety landing gears** to deploy, providing additional cushioning for a smooth landing.

To manage these operations efficiently, the **Parallax Propeller** processes input data from the IMU and ultrasonic sensors in real-time. The parachute mechanism works as a **primary safety measure**, significantly reducing the descent speed, while the safety landing gears acts as a secondary impact absorber**.** The ultrasonic sensors continuously relay distance measurements to the microcontroller, ensuring the timely activation of the landing gears deployment system. ensuring minimal impact damage while preventing unnecessary activations.

The entire setup is powered by a **2S 11.1V LiPo battery**, which supplies energy to the **drone's flight components** (motors, ESC, receiver-transmitter) while a separate **5V regulated power source** ensures stable operation of the propeller and sensors. The drone is designed to operate autonomously without pilot intervention in emergency conditions, making it a **reliable solution for commercial and research UAV applications**. Testing involves controlled drone flights where **various failure scenarios** (e.g., motor failure, signal loss, sudden tilt) are simulated to assess the **response time and effectiveness** of the parachute-balloon deployment system. The methodology ensures that the drone not only survives crashes but also allows for **easy recovery**, reducing the overall risk associated with aerial operations.

Workflow


Start
IMU Sensor Monitors Stability
Imbalance Detected?
Yes
Trigger Servo (Set 1)- Deploy Parachute
ultrasonic sensor measures Distance
reaches 30 cm
Landing Gears get depatched

Hardware Setup

The hardware was designed using Solidworks software. There are three main assemblies that are joined together:

- The Parachute assembly
- The drone Assembly
- The Ballon deployment assembly

The Parachute Assembly

The mainly consist of four plate- like structure which are topology optimized to reduce weight and maintain air flow preventing drag from the air pulled by the propellers.

Each plate is controlled by the servo horns which rotate 90 degrees to open the plate.

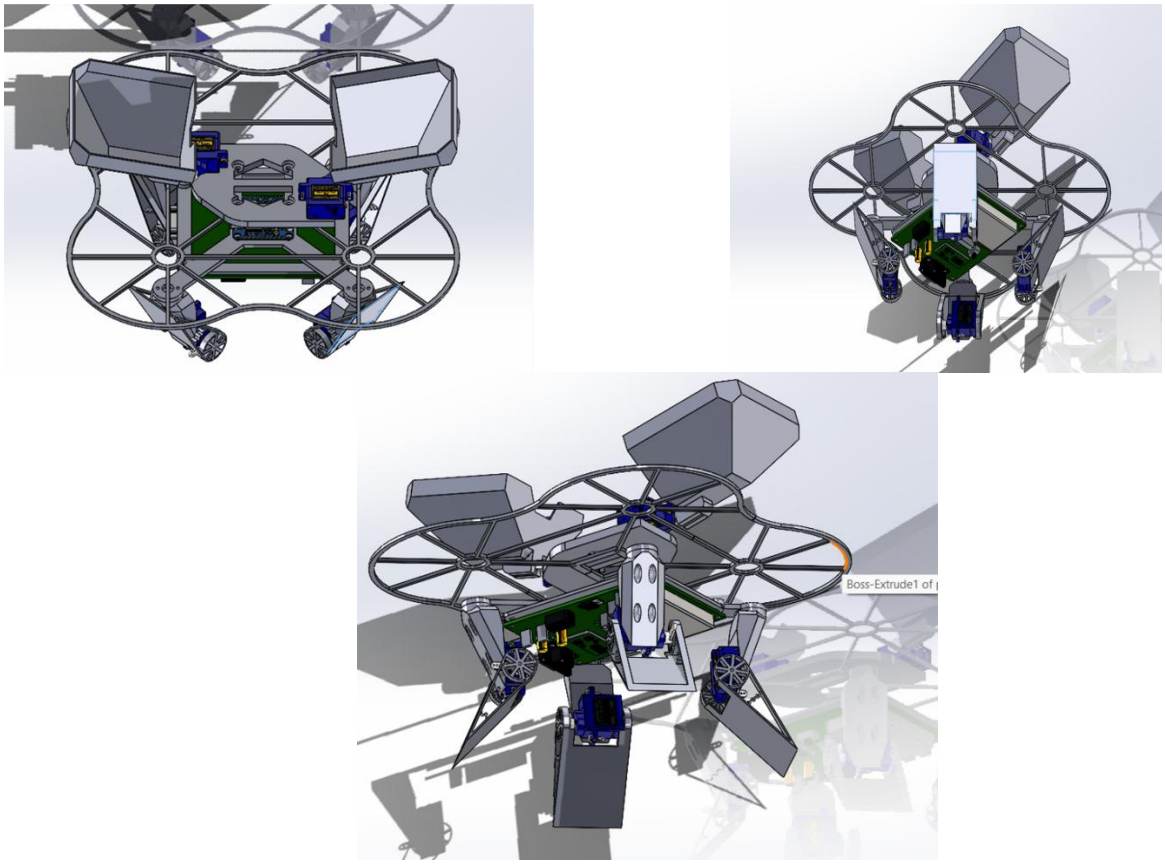


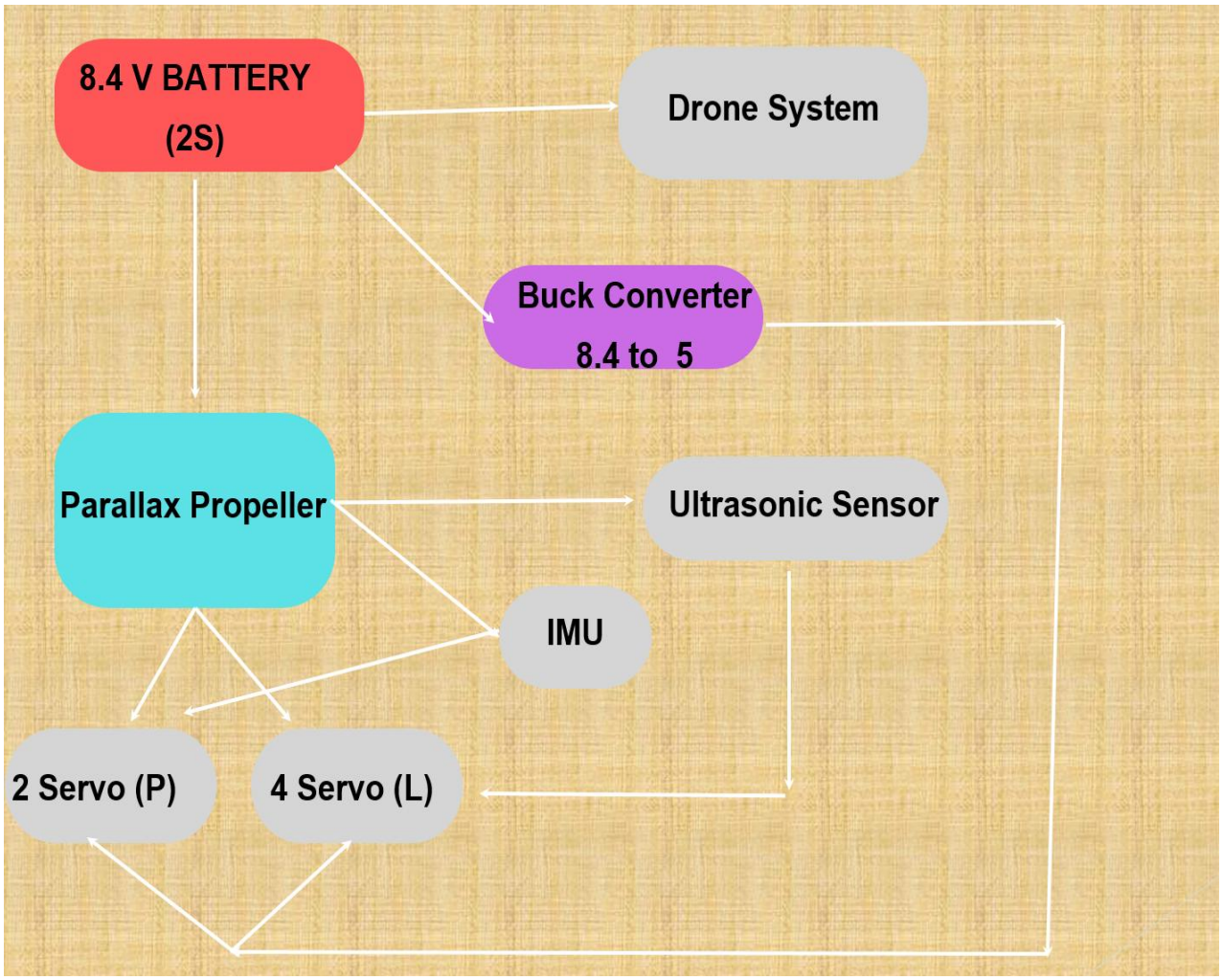
Fig 5: CAD Design

The Drone Assembly

A custom drone was designed to place the other to assemblies to the top and the bottom of the drone. The frame and the other parts are mainly made of PLA-CF (for high strength) and regular PLA.

Circuit Diagram

This circuit diagram outlines a Arduino-based system for sensing and control, for the parachute and balloon deployment system. The power supply, sensors, communication, and motor control are all carefully integrated to allow efficient operation.



Circuit Connection Overview

The circuit is built around an Arduino UNO R3, which acts as the main controller. The system is powered by a 12V battery, which is stepped down to 5V using a buck converter to provide the necessary operating voltage for the components.

Power Supply

1. A 8.4V battery is the main power source.
2. The buck converter steps down the voltage from 8.4 to 5V.
3. The 5V output is used to power the Servo motors.

Sensor and Peripheral Connections

The Propeller is interfaced with multiple peripherals, sensors, and communication modules as follows:

1. Ultrasonic Sensor
• Trig, Echo → P3

• This sensor measures distance using sound waves.
2. IMU (Inertial Measurement Unit)
• SDA → P0
• SCL → P1
• D0 → Gnd

• The IMU helps with motion sensing for orientation and stabilization.
3. Servo Motors
• 4 Servos (B) → P4
• 2 Servo (P) → P5
• These servos control mechanical components.

Drone Integration

• The Parallax Propeller also connects to a drone, with control inputs coming from the IMU and Ultrasonic Sensor.

Cost Analysis

Components	Cost
Parallax Propeller Board	$80
IMU Sensor	$7
Ultrasonic Sensor	$6
Servo Motors (x6)	$19
Drone Components	$200
Miscellaneous	$20
Total Estimated Cost	**$332**

Result

The prototype successfully detects imbalance, deploys the parachute, opens the landing gear, and offers minimum drone damage. Controlled tests show a significant reduction in impact force upon landing.

Future Scope

As drone technology continues to evolve, integrating advanced safety features will be crucial for their widespread adoption. This project lays the foundation for future enhancements, such as AI-based predictive stability control, which could analyze flight patterns in real-time and preemptively deploy safety measures before a critical failure occurs. Machine learning models can also be trained to optimize parachute deployment timing and safety balloon inflation based on environmental conditions.

Further advancements could include the integration of alternative landing mechanisms, such as deployable air cushions or retractable landing gears with shock absorbers. The system can also be expanded for industrial applications, where drones operate in extreme conditions such as high winds, remote terrains, or hazardous environments. Enhancements in battery efficiency and power management will ensure that safety systems operate reliably without compromising flight performance.

This project has significant potential in commercial applications, including drone-based deliveries, emergency response systems, and military UAV operations. As regulatory bodies enforce stricter safety guidelines for autonomous drones, such an intelligent landing system could become a mandatory feature, ensuring compliance while increasing drone longevity and reliability. Future research can explore the use of biodegradable parachute materials and compact gas-inflation mechanisms to further improve sustainability and efficiency.

Conclusion

The proposed system provides a reliable and cost-effective solution for drone safety. By combining IMU-based stability detection, ultrasonic altitude monitoring, parachute deployment, and landing gears, the system significantly reduces landing risks.

Reference

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5. UAV Regulations and Safety Guidelines - https://www.faa.gov/uas
6. Related Open-Source Drone Projects - https://github.com/PX4/PX4-Autopilot
7. Books on UAV Technologies and Control Systems - Various academic sources and publications.